ECLS/ECMO

Development of a Rolled Human-Scale Microfluidic Artificial Lung



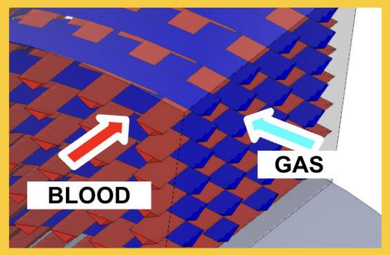
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Background

- Fabricate microfluidic artificial lungs with gas and blood channels separated by gas-permeable PDMS for effective CO_2 and O_2 exchange
- Improves oxygenation of blood from 70% to 95% saturation at 3 mL/min
- Offers a scalable and automated solution for producing large surface area microfluidic devices
- Clinical and commercial applications such as membrane-based gas exchange

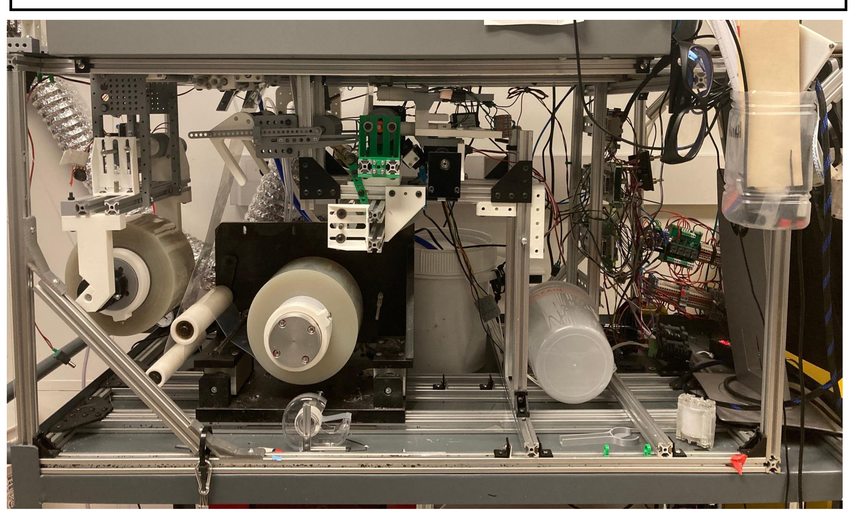
Alternating layers of blood and gas channels allow O_2 and CO_2 diffusion through the PDMS walls





Introduction

- Testing the Rolled-to-Rolled lung machine consistency/process
- Enhance centrifuge sensors



Methods

Procedure

- Core making process and Rolled-to-rolled machine

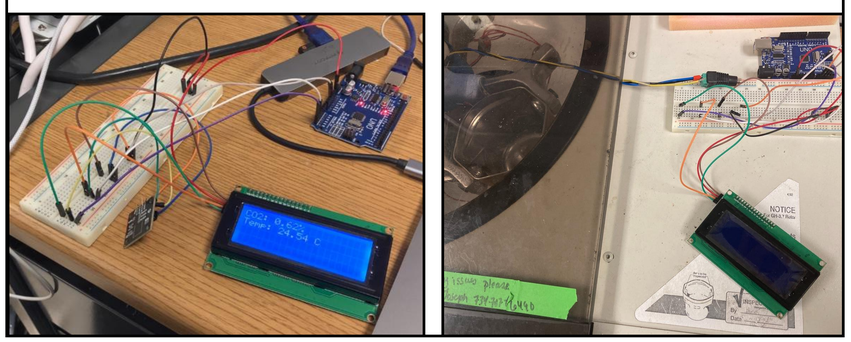
Circuit Design and Data collection

- Centrifuge temperature sensor for ensuring silicon A and B are mixed in the right conditions.

Wrinkles in core — **Optimized Roll**



Circuit Design



Humidity & CO_2 Data

Live Temperature and CO2 Data

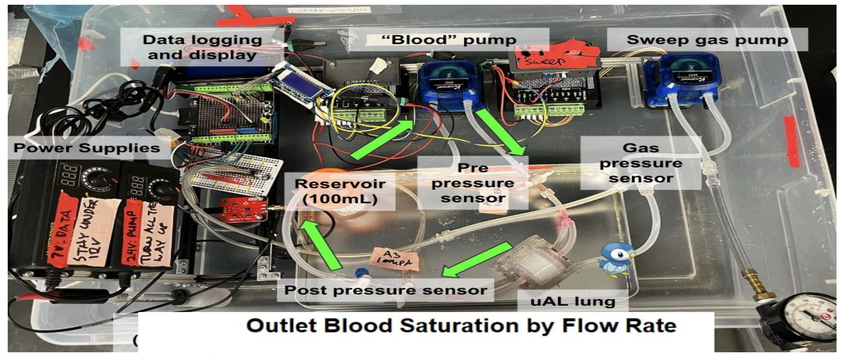
CO2 Concentration: **0.11 ppm**

PPM (parts per million) means the number of CO2 molecules for every 1 million molecules of air.

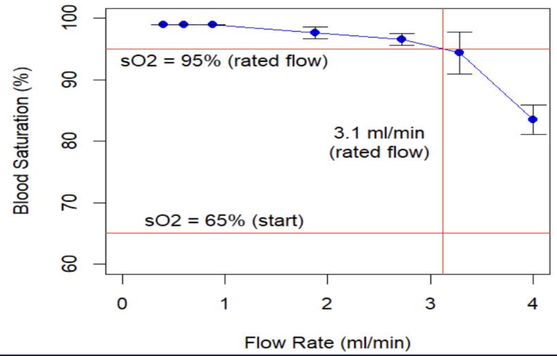
Temperature: **20.25 °C**

Tests

- Lung had 25 layers of engravings (13 gas and 12 blood)
- Test flows using log sequences of standard resistor values for 3-4 mins
- Maximum flow with >=95% SO2 estimated at 22mL/min
- Saw major decrease in oxygenation capabilities
- Hopefully preventable with more engraving layers



Outlet Blood Saturation by Flow Rate



Conclusion

- Consistency and optimization in manufacturing requires extensive research and trials
- Working on fixing R2R machine motors and electronics.
- Our team is still planning on testing this with >50 engraving layers

References

Zhang, A., Tharwani, K., Wang, J., Seilo, G. K., Atie, M. A., & Potkay, J. A. (2024). Roll-to-roll manufacturing of large surface area PDMS devices, and application to a microfluidic artificial lung. Lab on a chip, 24(18), 4357–4370. https://doi.org/10.1039/d4lc00339j